<u>Aether AI : AI Based SAAS Platform</u>

A

Project Report

Submitted for the partial fulfilment

of B.Tech. Degree

in

COMPUTER SCIENCE & ENGINEERING

by

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Declaration

We hereby declare that this submission is our own work and that, to the best of our belief and knowledge, it contains no material previously published or written by another person or material which to a substantial error has been accepted for the award of any degree or diploma of university or other institute of higher learning, except where the acknowledgement has been made in the text. The project has not been submitted by us at any other institute for requirement of any other degree.

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Certificate

This is to certify that the project report entitled “Aether AI+ : AI-Based SAAS Platform built using the PERN stack (PostgressSQL,Express.js, React, Node.js)” presented by Hamza Mubin, Rohan Kumar Pal, and Sharad Rajvanshi in the partial fulfilment for the award of Bachelor of Technology in Computer Science and Engineering, is a record of work carried out by them under my supervision and guidance at the department of Computer Science and Engineering at Institute of Engineering and Technology, Lucknow.

It is also certified that this project has not been submitted at any other Institute for the award of any other degrees to the best of my knowledge.

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Abstract

Artificial Intelligence is rapidly transforming the way people create digital content and use online services. However, many users still rely on multiple platforms for tasks such as content writing, image generation, background removal, and other creative activities, which can be time-consuming and inefficient. Keeping these challenges in mind, this project presents an AI-based SaaS platform that integrates multiple generative AI tools into a single user-friendly system.

The platform is designed as a full-stack web application where users can access various AI-powered features such as blog and article generation, AI image creation, background removal, text enhancement, and smart content assistance. The system aims to simplify creative workflows and improve productivity by providing fast, accessible, and automated AI solutions through a centralized platform.

The application provides secure user authentication, personalized dashboards, cloud-based media storage, subscription-based access, and responsive user interfaces for smooth user experience across devices. Generative AI models are integrated to produce human-like written content, generate creative visuals from text prompts, and process images efficiently with minimal user effort.

The main objective of this project is to create a smart and scalable SaaS platform that makes advanced AI tools easily accessible to students, creators, freelancers, businesses, and general users. By combining modern web technologies such as the PERN stack (PostgreSQL, Express.js, React.js, Node.js) with Generative AI APIs and cloud services, the proposed system attempts to deliver an efficient, interactive, and innovative digital content creation environment.

Keywords: Artificial Intelligence, Generative AI, SaaS Platform, PERN Stack, Blog Generation, Image Generation, Background Remover, React.js, Node.js, PostgreSQL, Cloud Storage, Content Creation

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Chapter 1. Introduction

1.1 Background and Motivation

Artificial Intelligence has become one of the fastest-growing technologies in recent years, transforming the way people create content, manage digital tasks, and interact with online platforms. Today, users expect smarter tools that can generate high-quality content instantly, automate repetitive tasks, and improve productivity with minimal effort. Despite the rapid growth of AI technologies, many users still depend on multiple separate applications for tasks such as blog writing, image generation, background removal, and content editing, making the overall workflow time-consuming and less efficient.

At the same time, Generative AI has emerged as a powerful solution capable of producing human-like text, realistic images, and intelligent digital assistance. AI-powered tools are now being widely used by students, content creators, freelancers, marketers, and businesses to simplify creative work and increase efficiency. However, many existing platforms either focus on only one feature or provide complex interfaces that are not easily accessible to general users.

The idea behind this project came from combining multiple AI-powered services into a single integrated SaaS platform. Instead of developing only a blog-writing tool or only an image-generation system, this project aims to provide a complete AI-based content creation environment where users can access several Generative AI features in one place. The platform allows users to generate blogs and articles, create AI-based images using text prompts, remove image backgrounds, and manage digital content efficiently through a modern and user-friendly interface.

The system is developed using the PERN stack, which includes PostgreSQL, Express.js, React.js, and Node.js, along with various Generative AI APIs and cloud-based services. The platform focuses on scalability, responsiveness, secure user authentication, and smooth user experience. By combining modern web technologies with artificial intelligence, the proposed system attempts to create a smart, accessible, and efficient SaaS platform for digital content creation and AI-assisted productivity.

1.2 Overview of the Proposed System

The proposed system is an AI-based SaaS platform designed to provide multiple Generative AI tools in one place. The platform allows users to register, log in, generate blogs and articles, create AI-based images from text prompts, remove image backgrounds, and manage their content through a simple and user-friendly dashboard.

The system also includes an admin panel for managing users, subscriptions, and platform activities. The backend handles authentication, database operations, API communication, and cloud storage integration to ensure secure and smooth performance.

1.3 Scope of Generative AI features and Technology Stack

The platform provides multiple AI-powered features such as blog and article generation, AI image creation from text prompts, background removal, and smart content enhancement. Users can quickly generate creative content, edit images, and manage their projects through a simple and interactive interface. The system is designed to support students, content creators, freelancers, and businesses by reducing manual effort and improving productivity.

The frontend of the application is developed using React.js, Vite, Tailwind CSS, and Axios to create a responsive and user-friendly interface. The backend is built using Node.js and Express.js, while PostgreSQL is used for secure and efficient database management. Additional cloud services are integrated for media storage, authentication, and subscription handling.

The platform uses Generative AI APIs and image-processing tools to generate high-quality text and visual content. The complete system is developed using the PERN stack, ensuring scalability, fast performance, and a smooth user experience across different devices.

1.4 Design Considerations and Usability

One important aspect of this project is the clear separation of responsibilities between users and administrators. Users can access AI tools such as blog writing, image generation, and background removal, while administrators manage users, subscriptions, and overall platform activities. This structure makes the platform organized, scalable, and easier to manage.

Another major focus of the project is usability. Many AI platforms become difficult to use because of complex interfaces and technical workflows. In this project, efforts have been made to keep the interface simple, responsive, and user-friendly so that even users with limited technical knowledge can use the platform comfortably.

The project also demonstrates how Generative AI can be integrated into a real-world full-stack application instead of being treated only as a separate experimental tool. By combining multiple AI-powered services into a single SaaS platform, the system provides a more practical, interactive, and efficient content creation experience for users.

1.5 Significance of the Project

The proposed AI-based SaaS platform is designed to bring multiple Generative AI services together into a single, accessible, and user-friendly environment. In today's digital world, content creators, students, freelancers, marketers, and businesses often depend on different platforms for tasks such as writing blogs, generating images, editing media, and improving content quality. This project attempts to simplify that process by providing all these AI-powered services under one platform, making digital content creation faster, easier, and more efficient.

One of the major significances of the project is its practical use of Generative AI in real-world applications. Instead of limiting artificial intelligence to experimental or research-based systems, this platform demonstrates how AI can directly assist users in their daily creative and professional tasks. Features such as AI blog generation, image creation from text prompts, and background removal help reduce manual effort and save time while improving productivity and creativity.

From a technical perspective, the project demonstrates the successful integration of modern web technologies with cloud-based AI services. The platform is developed using the PERN stack, which includes PostgreSQL, Express.js, React.js, and Node.js, providing a scalable, responsive, and efficient full-stack architecture. The project also incorporates JWT-based authentication, API integration, cloud storage services, and subscription management, making the implementation closer to real-world SaaS applications used in the industry.

Another important aspect of the project is usability and accessibility. Many AI tools available today require technical knowledge or involve complicated workflows. This platform focuses on creating a clean and simple user interface so that even non-technical users can access advanced AI features without difficulty. By keeping the system user-friendly and responsive, the platform becomes more practical for everyday use across different devices and user groups.

The project also highlights the growing importance of AI-powered automation in digital content creation. As businesses and individuals increasingly rely on online content for communication, marketing, branding, and education, tools that can generate high-quality content quickly have become highly valuable. This platform demonstrates how Generative AI can support these needs by producing creative text and visual content efficiently.

From an educational perspective, the project serves as a complete learning experience covering different stages of software development. It includes requirement analysis, system design, frontend and backend development, database management, API integration, cloud services, authentication, deployment, and AI integration. Working on such a project helps in understanding both theoretical concepts and their practical implementation in real-world environments.

Overall, the project reflects how modern web technologies and Generative AI can work together to build intelligent, scalable, and user-centric applications that improve productivity, simplify workflows, and enhance the digital experience for users.

Chapter 2. Literature Review

2.1 Existing AI-Based SaaS Platforms and Generative AI Research

The field of Artificial Intelligence has experienced rapid growth in recent years, especially with the rise of Generative AI and cloud-based SaaS (Software as a Service) platforms. Businesses, content creators, students, and developers are increasingly adopting AI-powered applications to automate repetitive tasks, improve productivity, and simplify digital content creation. As a result, modern AI platforms now provide services such as content writing, image generation, background removal, resume analysis, and intelligent automation through web-based applications.

Several AI-powered SaaS platforms already exist that provide content generation and media processing services. Many of these systems focus on specific functionalities such as AI article writing, image editing, chatbot assistance, or text-to-image generation. These platforms help users reduce manual work and generate creative content more efficiently. However, most existing systems are designed around a single feature or service, requiring users to switch between multiple applications for different tasks.

At the same time, research in Generative AI has expanded significantly. Large Language Models (LLMs) and AI image-generation systems have shown impressive performance in generating human-like text, realistic images, and intelligent responses from user prompts. Technologies such as transformer-based language models, text-to-image diffusion models, and AI-powered image-processing tools are now widely used in digital platforms. These technologies have made it possible to automate tasks like blog writing, article generation, resume review, image creation, and content enhancement with high accuracy and speed.

Many existing AI content-generation systems mainly focus on text-based features such as article writing or chatbot interaction. Similarly, some AI tools specialize only in image generation or image editing functionalities. Although these applications provide useful services, users often need separate tools for different creative tasks, which reduces efficiency and creates workflow fragmentation. In practical scenarios, users generally prefer a single platform where multiple AI-powered services can be accessed together through one interface.

Another issue observed in many existing AI platforms is the complexity of their interfaces and workflows. Some systems are designed mainly for developers or advanced users, making them difficult for non-technical users to operate. In many cases, users also face limitations related to content storage, subscription management, authentication, or cloud accessibility. These challenges reduce the overall usability and scalability of such platforms.

Recent developments in cloud computing and AI integrations have improved the performance and accessibility of SaaS applications. Modern platforms now use cloud-based databases, secure authentication systems, cloud media storage, and third-party AI APIs to provide scalable and efficient services. AI-powered applications are increasingly integrating technologies such as Google Gemini API, image-processing APIs, and cloud storage platforms to improve functionality and user experience.

Researchers and developers have also focused on integrating multiple AI services into unified SaaS environments. Combining text generation, image creation, background removal, and media management into a single platform improves usability and creates a smoother workflow for users. Such integrations make AI applications more practical for real-world use and reduce dependency on multiple standalone tools.

The proposed project, *AetherAI*, follows this integrated approach by combining multiple Generative AI features into one full-stack SaaS platform. The system provides AI article generation, blog title creation, AI image generation, resume review, and background removal through a modern and responsive interface. The platform is developed using modern web technologies and cloud-based AI services to provide a scalable, secure, and user-friendly experience.

From a technical perspective, the project demonstrates the practical integration of the PERN stack with third-party Generative AI APIs and cloud services. Technologies such as React.js, TypeScript, Node.js, Express.js, PostgreSQL, Clerk authentication, Cloudinary storage, and AI APIs are combined to build a real-world AI-powered SaaS application. This shows how modern web development and Generative AI can work together to create intelligent and scalable digital platforms for content creation and productivity enhancement.

2.2 Research Gap and Proposed Approach

During the study of existing AI platforms and Generative AI applications, it was observed that most systems focus only on a single functionality such as blog writing, image generation, resume analysis, or background removal. Although these tools perform well in their specific areas, users often need to switch between multiple platforms to complete different creative tasks. This creates workflow fragmentation, increases complexity, and reduces overall productivity. This observation became one of the major motivations behind the proposed project.

The proposed system, *AetherAI*, attempts to bridge this gap by integrating multiple Generative AI services into a single unified SaaS platform. Instead of limiting the application to only one AI feature, the system combines AI article generation, blog title generation, resume review, AI image generation, and background removal within the same ecosystem. This approach provides users with a smoother and more efficient workflow through a centralized platform.

Another important aspect considered during the development of this project is usability. Many existing AI platforms become difficult to use because of complicated interfaces, technical settings, or developer-focused workflows. In this project, efforts have been made to keep the interface simple, responsive, and user-friendly so that even non-technical users can access AI-powered features without difficulty.

The system also differs from many experimental AI projects because it integrates Generative AI functionalities into a complete real-world full-stack SaaS application rather than limiting them to standalone scripts or isolated demonstrations. Features such as secure authentication, subscription management, cloud-based image storage, responsive dashboards, and AI API integrations make the platform more practical and closer to industry-level SaaS applications.

Another gap identified during the literature review was the lack of unified AI-powered content creation environments. Many existing systems provide text-generation tools separately from image-processing tools, forcing users to rely on multiple applications. The proposed platform addresses this issue by combining text-based AI services and image-processing functionalities within a single responsive interface, improving accessibility and convenience for users.

The literature survey also revealed that several AI platforms focus mainly on functionality while giving less attention to scalability, user experience, and cloud integration. The proposed system addresses these limitations by using modern technologies such as the PERN stack, cloud-based database management, secure authentication systems, and cloud media storage to create a scalable and efficient architecture.

Overall, the literature review confirmed that while individual AI services such as article generation, AI image creation, and background removal already exist separately, a unified AI-powered SaaS platform that combines all these features within a modern full-stack application remains relatively less explored at the student project level. This observation directly influenced the design, scope, and implementation of the proposed *AetherAI* platform.

Chapter 3. Methodology

Chapter 3: Methodology

The development of *Aether AI* involved combining modern web technologies with Generative AI services to create a smart and user-friendly SaaS platform. Since the platform includes multiple AI-powered features such as blog writing, AI image generation, resume review, and background removal, the implementation was divided into separate modules that work together within a unified system.

The overall methodology focuses on three major objectives:

- providing an easy-to-use AI-powered content creation platform,
- integrating multiple Generative AI services into one system,
- and ensuring smooth communication between the frontend, backend, database, and AI APIs.

The project follows a modular architecture where the frontend application, backend APIs, database, authentication system, and AI services operate independently but remain connected through APIs and shared workflows.

3.1 System Overview

The proposed system is divided into four major parts:

- User-side web application
- Admin dashboard and subscription management
- Backend server and database
- Generative AI integration module

Each component performs a specific role within the platform.

The user-side application allows users to register, log in, generate AI articles, create blog titles, generate images from text prompts, remove image backgrounds, upload resumes for review, and manage generated content through a personalized dashboard.

The admin dashboard is responsible for managing users, monitoring subscriptions, handling platform activities, and maintaining overall system performance.

The backend acts as the central communication layer between the frontend, database, cloud storage services, authentication system, and third-party AI APIs. It handles request processing, authentication, API communication, content storage, and secure data management.

The Generative AI module processes user requests using integrated AI services for text generation, image creation, resume analysis, and image processing. AI-generated outputs are then stored and displayed through the platform interface.

The overall architecture helps keep the project scalable, organized, and efficient because each module can function independently while still working together as part of a complete AI-powered SaaS platform.

3.2 Frontend Development

The frontend of the project has been developed using React.js, TypeScript, and Vite. React was selected because it supports reusable component-based development and provides a smooth and interactive user experience through dynamic rendering. TypeScript improves code scalability and type safety, while Vite was used to increase development speed and optimize application performance.

Tailwind CSS was used for designing the user interface. Instead of relying heavily on traditional CSS files, utility-based styling was used to create responsive layouts, modern designs, and consistent UI components across the platform.

The frontend is divided into two main sections:

3.2.1 User Application

The user-side application provides access to all AI-powered services available on the platform. Important functionalities include:

- User registration and login
- AI article generation
- AI blog title generation
- AI image generation from text prompts
- AI background removal
- Resume upload and AI-based resume review

- Content history and dashboard management
- Subscription and plan access

React Router DOM was used for page navigation, while Axios was used for communication with backend APIs.

The platform allows users to interact with different Generative AI tools through a simple and responsive interface. Generated content and images are displayed dynamically and stored securely for future access.

Toast notifications were integrated to provide real-time feedback messages such as successful login, content generation status, image processing completion, and upload confirmations.

3.2.2 Admin Dashboard

A separate admin dashboard was developed to manage platform activities and monitor user operations. This separation improves security and keeps administrative functionalities independent from normal user activities.

The admin dashboard includes features such as:

- Managing users and subscriptions
- Monitoring AI tool usage
- Viewing generated content statistics
- Managing premium feature access
- Monitoring platform performance and activities

Role-based authentication was implemented using JWT tokens and Clerk authentication to ensure that only authorized users can access protected routes and administrative functionalities.

3.3 Backend Development

The backend of the project was developed using Node.js and Express.js. The backend acts as the central communication layer of the platform and handles major operations such as authentication, AI API integration, database communication, subscription handling, media processing, and cloud storage management.

PostgreSQL (Neon) was used as the primary database because of its scalability, cloud-based architecture, and efficient data management capabilities. The database stores user information, generated articles, AI image URLs, resume review results, subscription details, and platform activity records.

The backend is organized into different API modules such as:

- User APIs
- AI Generation APIs
- Authentication APIs

- Admin APIs
- Subscription and Media APIs

These APIs are responsible for handling requests from the frontend application and managing communication between the platform and third-party AI services.

3.3.1 Authentication and Security

Secure authentication was implemented using Clerk authentication and JWT-based session management. When a user logs in successfully, secure session tokens are generated and validated to protect private routes and user data.

Role-based access control was also implemented to separate normal user operations from administrative functionalities. This ensures that only authorized users can access protected dashboards and premium platform features.

Middleware functions were used for request validation, authentication verification, subscription checking, and API protection. These security mechanisms help maintain a safe and reliable environment for users interacting with AI-powered services.

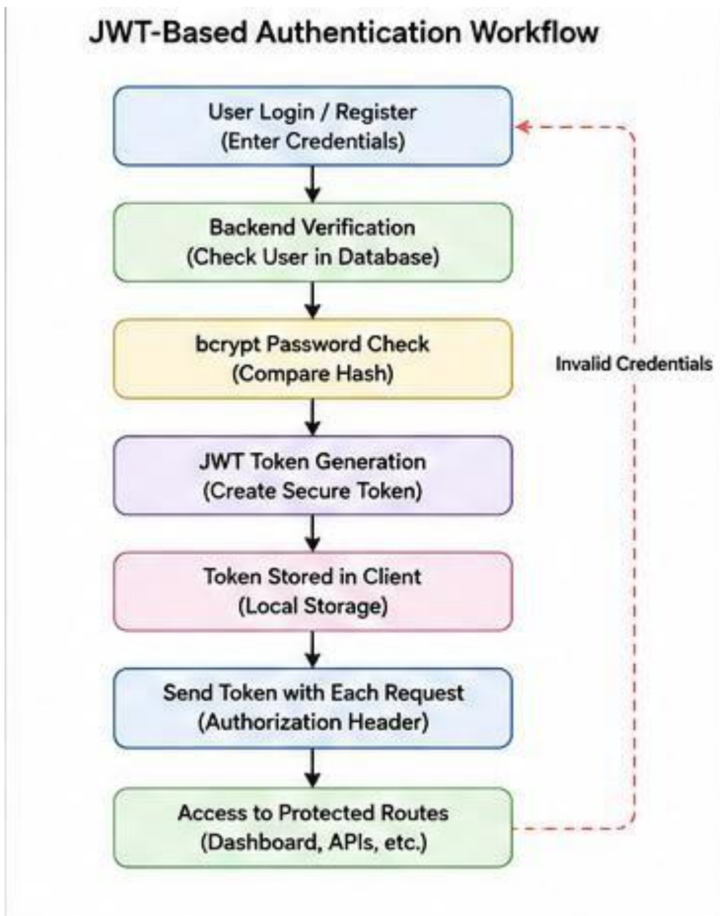

Figure 3.1: JWT-Based Authentication Workflow

3.3.2 AI Content and Media Management

The AI content and media management system is one of the central parts of the project. Users can generate AI-written articles, create blog titles, generate images from text prompts, upload resumes for review, and remove image backgrounds directly through the platform.

The backend processes user requests by communicating with third-party Generative AI APIs. Text-based requests such as article generation and resume analysis are handled through AI language models, while image-related functionalities such as text-to-image generation and background removal are processed using image AI services.

Generated content and processed images are securely stored in the database and cloud storage system for future access. The platform also maintains user activity records, generated content history, and subscription-based feature access.

Additionally, users can manage their generated content through the dashboard, while administrators can monitor AI tool usage and overall platform activities when required.

3.3.3 AI Subscription Payment and Verification Workflow

To support subscription plans and premium AI features, a secure payment gateway was integrated into the platform. The backend is responsible for generating payment orders, handling transaction requests, and verifying successful payments before activating premium access for users.

This integration ensures secure and reliable online transactions within the platform. After successful payment verification, users gain access to premium AI functionalities such as advanced content generation and additional usage limits.

The payment workflow is designed to provide a smooth and secure experience while maintaining proper transaction validation and subscription management.

3.3.4 Image Upload and Cloud Storage

The platform supports image upload and cloud storage functionalities for AI-generated images and processed media files. Image uploads are handled using Multer middleware, which manages multipart form data efficiently within the backend.

Instead of storing images on the local server, all generated and processed images are uploaded to Cloudinary for secure cloud-based storage. After image generation or background removal, the backend uploads the processed image to Cloudinary and stores the returned image URL in the PostgreSQL database.

Cloud-based storage helps reduce server load, improves scalability, and allows users to access their generated images securely from anywhere through the platform.

3.4 AI Image Generation and Processing Workflow

The Generative AI module was developed to provide multiple AI-powered functionalities within the platform. Different third-party AI services and APIs were integrated to handle text generation, image creation, resume analysis, and image-processing tasks efficiently.

The module supports features such as:

- AI article generation
- AI blog title generation
- AI resume review
- AI text-to-image generation
- AI background removal

Users provide prompts, text inputs, images, or resume files through the frontend interface, and the system processes these requests using integrated Generative AI APIs.

Different AI services were used for different functionalities, including:

- Google Gemini API for text-based AI generation
- ClipDrop API for image generation and background removal
- Cloudinary for cloud image storage and management

The backend communicates with these AI services through secure API requests and returns the generated results to the frontend interface in real time.

Generated articles, AI images, resume review results, and media URLs are stored in the PostgreSQL database for future access and content management. The modular architecture of the AI integration system makes the platform scalable and allows additional AI tools to be added easily in the future.

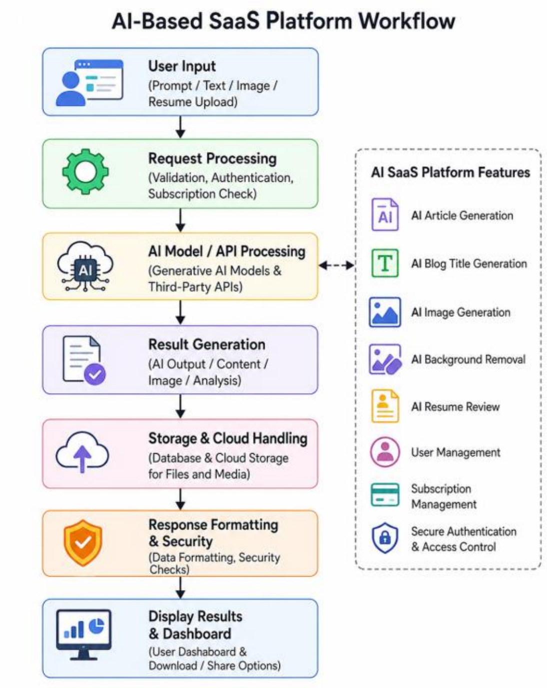

Figure 3.4: Workflow

3.5 Background Removal System Architecture

PostgreSQL database tables were created to manage platform data efficiently. The major database models include:

- User Model
- Generated Content Model
- AI Image Model
- Subscription Model
- Resume Review Model

The User Model stores user-related information such as profile details, authentication data, subscription plans, and activity history.

The Generated Content Model stores AI-generated articles, blog titles, prompts, timestamps, and publishing information.

The AI Image Model manages AI-generated images, background-removed images, image URLs, and cloud storage references.

The Subscription Model maintains premium plan details, payment status, feature access, and subscription validity.

The Resume Review Model stores uploaded resume information, extracted content, AI-generated feedback, and review history.

These database models work together to maintain smooth communication between users, AI services, cloud storage, and the overall SaaS platform workflow.

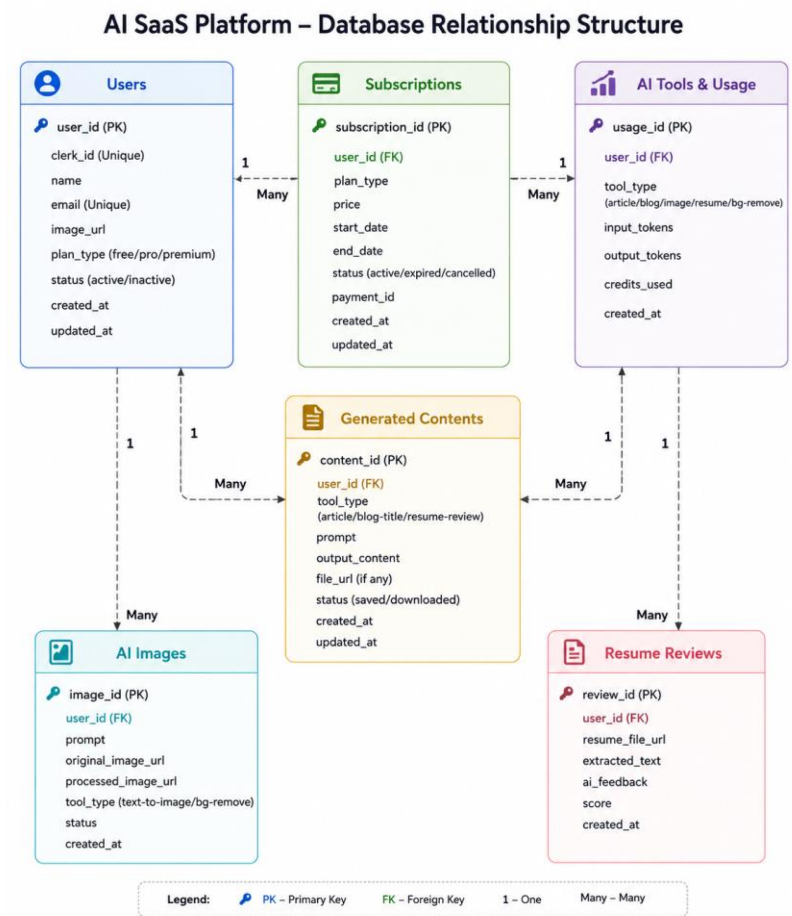

Figure 3.5: Database Relationship Structure

3.6 Database Relationship Structure (PostgreSQL)

The complete platform operates through interaction between the frontend application, backend server, PostgreSQL database, cloud storage services, and Generative AI APIs. To manage platform data efficiently, multiple PostgreSQL tables were created and connected through relational mappings.

The major database models used in the system include:

- User Model
- Subscription Model
- Generated Content Model

- AI Image Model
- Resume Review Model
- AI Tool Usage Model

The User Model stores user-related information such as profile details, email, authentication data, subscription plans, account status, and activity timestamps.

The Subscription Model manages premium plans, payment information, subscription validity, billing status, and user access control for premium AI features.

The Generated Content Model stores AI-generated articles, blog titles, prompts, generated outputs, publishing status, and content history. This allows users to access and manage previously generated content through the dashboard.

The AI Image Model stores information related to AI-generated images and background removal tasks. It maintains image prompts, original image references, processed image URLs, cloud storage links, and generation timestamps.

The Resume Review Model stores uploaded resume details, extracted text, AI-generated feedback, review scores, and processing history for resume analysis features.

The AI Tool Usage Model maintains records of tool usage, generated requests, token consumption, and user activity statistics. This helps in monitoring platform usage and subscription limitations.

The database relationships are designed using foreign key mappings to maintain proper connections between users, subscriptions, generated content, AI images, and review records. One user can generate multiple AI contents, images, and resume reviews, while subscription records control feature accessibility and premium usage.

This relational database structure improves scalability, maintains data consistency, and allows smooth communication between different modules of the AI SaaS platform. The PostgreSQL-based architecture also supports efficient query handling, secure data management, and real-time interaction with cloud services and AI APIs.

3.7 Tools and Technologies Used

The major technologies used during the development of the AI-based SaaS platform are listed below.

3.7.1 Frontend

- React.js
- TypeScript
- Vite
- Tailwind CSS
- Axios
- React Router DOM

React.js was used to build the interactive and responsive user interface of the platform. TypeScript improved scalability and type safety during development. Vite was selected as the frontend build tool because it provides faster development and optimized performance. Tailwind CSS was used for creating responsive and modern UI components, while Axios handled API communication between the frontend and backend. React Router DOM was used for navigation and route management within the application.

3.7.2 Backend

- Node.js
- Express.js
- PostgreSQL (Neon)
- JWT Authentication
- bcrypt
- Multer

Node.js and Express.js were used to develop the backend server and REST APIs. The backend manages authentication, AI API communication, database operations, cloud storage handling, and subscription management.

PostgreSQL (Neon) was used as the primary cloud-based relational database for storing user data, generated content, AI image records, subscription information, and activity logs.

JWT authentication was implemented for secure session management and protected route access. Passwords and sensitive data were secured using bcrypt encryption. Multer middleware was used to manage file uploads such as resume PDFs and images for background removal.

3.7.3 Generative AI and Processing Services

- Google Gemini API
- ClipDrop API
- pdf-parse
- sharp

Google Gemini API was integrated for text-based AI functionalities such as AI article generation, blog title generation, and resume review analysis.

ClipDrop API was used for image-based AI functionalities including text-to-image generation and AI background removal.

The pdf-parse library was used to extract readable text from uploaded resume PDFs before sending the content for AI analysis. The sharp library was used for image resizing and preprocessing before image generation or background removal operations.

3.7.4 External Services and Cloud Integration

- Cloudinary
- Clerk Authentication
- Neon PostgreSQL

Cloudinary is a cloud-based media management service used for storing and serving AI-generated and processed images. When images are generated or modified, the backend uploads them to Cloudinary and stores the secure image URL in the PostgreSQL database. This approach reduces server load and improves media accessibility through cloud storage.

Clerk Authentication was integrated for secure user authentication, role management, and subscription handling. It manages login sessions, user verification, and premium feature access within the platform.

Neon PostgreSQL provides scalable cloud-native database management for storing application data efficiently and securely.

The combination of these technologies helped create a responsive, scalable, and real-world AI SaaS platform capable of handling multiple Generative AI workflows efficiently.

Chapter 4. Experimental Results

The implementation of the proposed AI-based SaaS platform was carried out by integrating the frontend application, backend APIs, PostgreSQL database, cloud services, and Generative AI modules into a single working system. After completing the development phase, different functionalities of the platform were tested individually as well as collectively to ensure smooth interaction between all components.

The system was executed locally and tested using separate environments for the frontend application, backend server, database services, authentication system, and AI integrations. Different test cases were performed to verify authentication, AI content generation, image generation, background removal, resume review functionality, cloud storage handling, and dashboard operations.

The main objective during testing was not only to check whether the individual features were functioning correctly, but also to evaluate whether the overall platform provided a smooth and natural user experience. Since the system integrates multiple Generative AI services and cloud-based functionalities together, stable communication between the frontend, backend, database, and third-party AI APIs was essential for reliable performance.

Special attention was also given to response time, content generation accuracy, secure authentication, media processing, and subscription-based feature access. The testing phase confirmed that the platform was capable of handling different AI-powered workflows efficiently while maintaining responsiveness and usability across different modules.

4.1 User Authentication – Login Interface

The authentication system was tested for normal users, administrators, and subscription-based access separately. Users were successfully able to create accounts, log in to the platform, and access authorized features based on their authentication status and subscription plan.

JWT-based authentication and Clerk authentication were used to maintain secure sessions throughout the platform. Invalid login attempts, unauthorized dashboard access, expired sessions, and protected route access were also tested to verify the security mechanisms of the application.

The following functionalities were verified successfully:

- User registration
- User login and authentication
- Admin access control
- Protected route access
- JWT token verification
- Session management

- Logout functionality
- Subscription-based feature access

Password encryption and secure authentication mechanisms ensured that sensitive user information was protected and not exposed directly inside the database.

The testing results confirmed that the authentication system was stable, secure, and capable of handling user access efficiently across different modules of the AI SaaS platform.

4.2 AI Blog Writing Dashboard

The AI blog writing module was tested using multiple user accounts and different content prompts. Users were successfully able to:

- Enter blog topics and prompts
- Generate AI-written articles
- Create blog titles
- View generated content history
- Access saved AI-generated content

The AI generation workflow functioned correctly by processing user prompts and returning generated content in real time. Generated articles and titles were stored successfully in the database and reflected properly within the user dashboard.

Content generation speed, prompt handling, and dashboard responsiveness were also verified successfully.

Observations

- AI article generation worked smoothly without major delays.
- Blog titles were generated accurately based on user prompts.
- Generated content was stored and displayed correctly in the dashboard.
- Users were able to access previous content history successfully.
- The interface remained responsive during consecutive AI requests.

The system behaved consistently even when multiple content-generation requests were tested continuously, confirming stable communication between the frontend, backend, database, and Generative AI APIs.

4.3 AI Image Generation Using Text Prompts

The AI image generation module was tested using different user accounts and various text prompts. Users were successfully able to:

- Enter text prompts for image generation
- Generate AI-based images
- View generated image results
- Save generated images
- Access image history through the dashboard

The system successfully processed text prompts through the integrated AI image-generation API and returned generated images in real time. Generated images were uploaded to Cloudinary cloud storage, and the image URLs were stored securely in the PostgreSQL database.

The dashboard displayed:

- Generated image previews
- Prompt history
- Image generation timestamps
- Saved image records

Testing also confirmed that users could:

- Generate multiple images consecutively
- Access previously generated images
- Download or view generated content
- Use the feature through a responsive interface

The separation between normal users and premium-access users functioned correctly, ensuring that advanced AI image-generation features remained accessible only to authorized users based on their subscription plan.

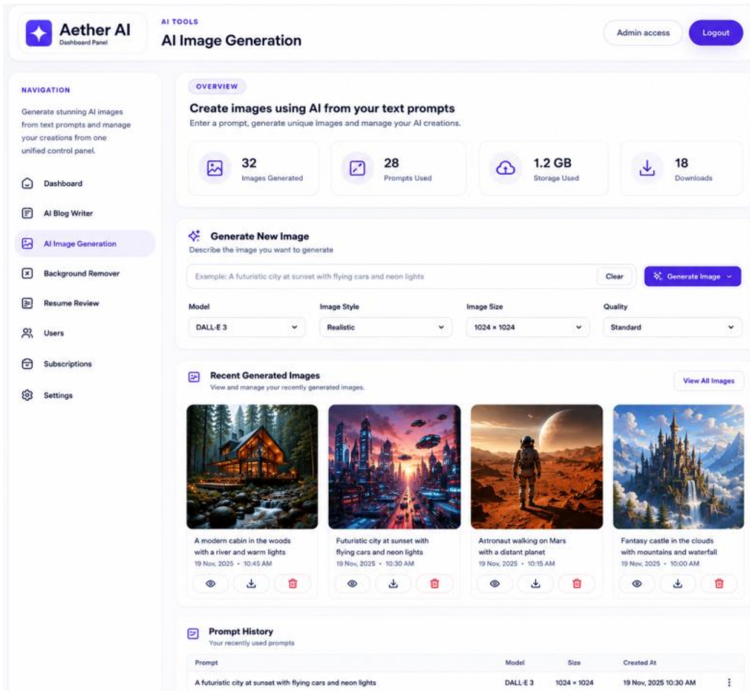

Figure 4.3: Image generation

4.4 Background Remover Output Screen

The AI background removal module was tested using multiple image formats and different user accounts. The background removal workflow included:

- Uploading images through the frontend interface
- Processing images using the AI background removal API
- Generating background-removed outputs
- Uploading processed images to cloud storage
- Displaying final output results in the dashboard

Test image-processing operations were completed successfully during implementation. During testing, users were able to upload images through both desktop and mobile browsers without major issues. The system processed uploaded images correctly and generated background-removed outputs within a reasonable response time.

After successful processing, the generated images were automatically uploaded to Cloudinary cloud storage, and secure image URLs were stored in the PostgreSQL database. The processed images appeared instantly within the user dashboard and could be viewed or downloaded successfully.

Results Observed

- Image uploads were processed successfully.
- Background removal results were generated correctly.
- Processed images were stored securely in cloud storage.
- Generated image URLs worked properly across the platform.
- Download and preview functionality operated smoothly.
- Multiple image formats such as JPEG, PNG, and WebP were supported successfully.

The Cloudinary integration automatically optimized processed images for faster delivery and reduced loading time during testing. The responsive interface also ensured that the background remover feature remained easy to use even for non-technical users.

Overall, the background removal module performed consistently and demonstrated stable interaction between the frontend interface, backend APIs, AI processing services, database, and cloud storage system.

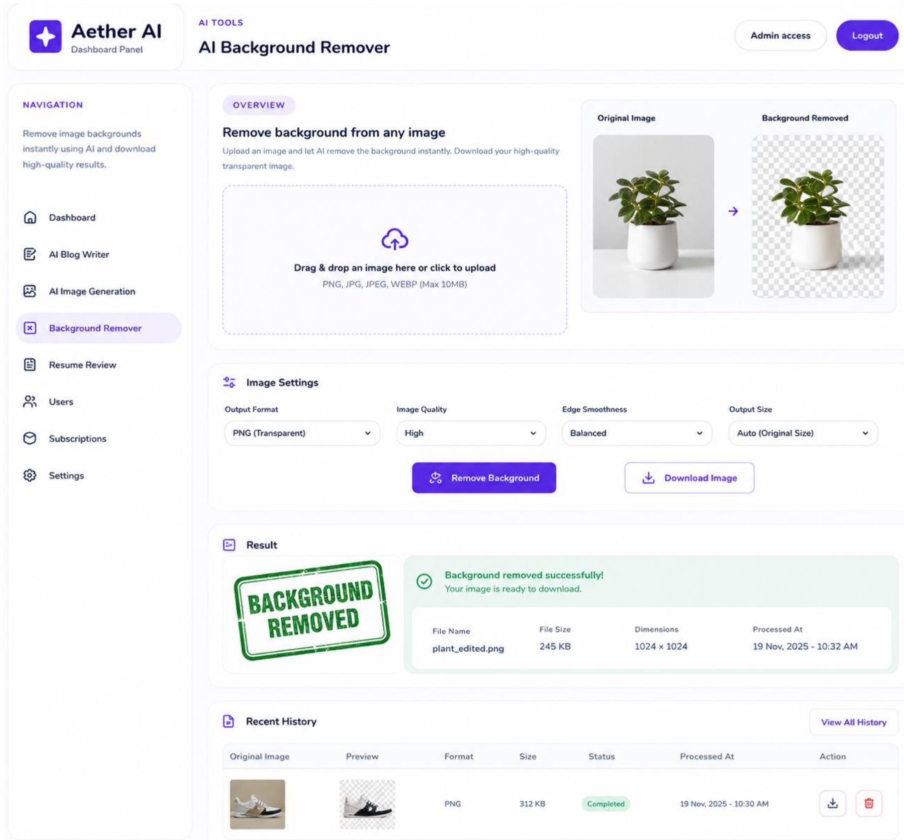

Figure 4.4: Background Removed

4.5 User Dashboard with AI Tools Access.

The user dashboard was designed to provide centralized access to all AI-powered tools available within the *Aether AI* platform. The dashboard allowed users to navigate between different Generative AI features through a clean and responsive interface.

The dashboard provided access to functionalities such as:

- AI article generation
- AI blog title generation
- AI image generation
- AI background removal
- AI resume review
- Subscription and account management

During testing, users were able to switch smoothly between different AI tools without performance issues. The sidebar navigation and dashboard layout helped users access features quickly and manage generated content efficiently.

The dashboard also displayed important information such as:

- Recent AI-generated content
- Generated image history
- User subscription status
- Tool usage activity
- Saved projects and downloads

Users were successfully able to generate, view, download, and manage AI-generated outputs directly from the dashboard interface. The responsive design ensured smooth usability across both desktop and mobile devices.

Results Observed

- Navigation between AI tools worked smoothly.
- Generated content history updated correctly.
- Dashboard statistics displayed accurately.
- Subscription-based access control functioned properly.
- User interface remained responsive during continuous usage.

The integration of multiple AI services within a single dashboard improved usability and created a more organized workflow for users. Overall, the dashboard successfully acted as the central control panel for accessing and managing all AI-powered services within the *Aether AI* platform.

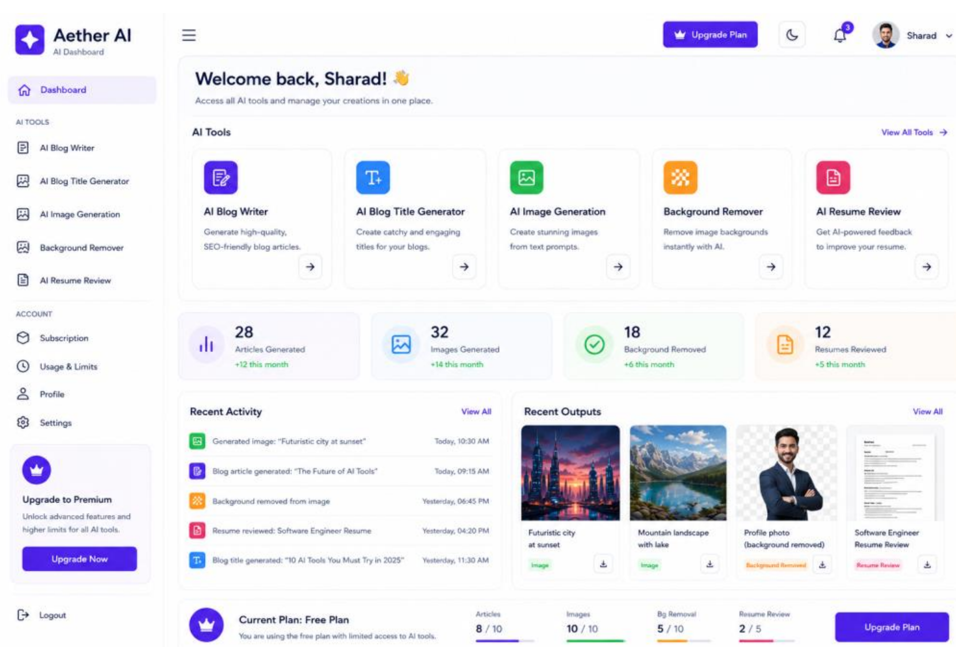

Figure 4.5: User Dashboard with AI Tools Access.

4.6 Content History and Saved Projects

The content history and saved projects module was tested to verify whether users could access, manage, and organize their previously generated AI outputs efficiently. The platform successfully stored generated articles, blog titles, AI-generated images, background-removed images, and resume review results within the user dashboard.

Users were able to:

- View previously generated AI content
- Access saved projects and outputs
- Download generated images and documents
- Revisit earlier prompts and generated results
- Manage AI-generated content history

The dashboard displayed detailed information related to generated projects, including:

- Content title or prompt
- Generation date and time
- Tool type used
- Download and preview options
- Processing status

During testing, generated outputs were retrieved successfully from the PostgreSQL database and displayed dynamically within the dashboard interface. AI-generated images and processed media files stored in Cloudinary were also accessible through secure URLs without loading issues.

Results Observed

- Saved projects loaded correctly from the database.
- Generated content history updated automatically.
- Download and preview functionality worked properly.
- AI-generated outputs remained accessible after multiple sessions.
- Dashboard performance remained stable even with multiple saved records.

The content management system improved the usability of the platform by allowing users to organize and revisit their previous AI-generated work without regenerating content repeatedly. Overall, the feature provided a smoother and more practical user experience within the *Aether AI* platform.

4.7 Subscription and Payment Management Interface

The subscription and payment management system was tested to verify secure plan activation, payment verification, and premium feature access within the *Aether AI* platform. Users were successfully able to upgrade plans, access premium AI tools, and manage subscription details through the dashboard interface.

The payment workflow included:

- Selecting a subscription plan
- Creating a secure payment order
- Redirecting users to the payment interface
- Verifying transaction details
- Activating premium access after successful payment

During testing, the payment interface opened correctly on both desktop and mobile browsers. After successful transactions, subscription status and premium feature access were updated automatically within the user dashboard.

The dashboard successfully displayed:

- Current subscription plan
- Usage limits and remaining credits
- Payment history
- Premium feature access
- Subscription validity details

Results Observed

- Payment requests were generated successfully.
- Subscription activation worked correctly after payment verification.
- Premium AI tools became accessible instantly after upgrade.
- Failed transactions did not activate premium features.
- Usage statistics and subscription details updated dynamically.

The subscription management interface improved the practical usability of the platform by providing a smooth and secure method for handling premium AI services and user access control.

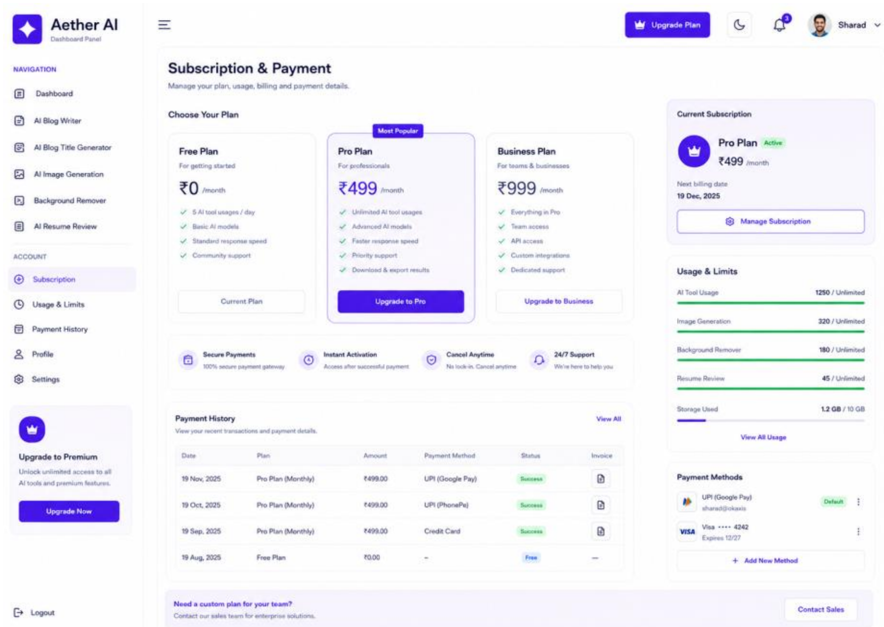

Figure 4.7: Subscription and Payment Management Interface

4.8 AI-Generated Blog Content Result

Different AI-generated content prompts were tested during the development of the blog writing module. The system was evaluated using multiple content categories such as technology, education, business, marketing, and artificial intelligence topics.

The AI blog writing module successfully generated:

- Full-length blog articles
- SEO-friendly content

- Blog titles and headings
- Structured paragraphs
- Creative and informative text outputs

Various prompt styles and content lengths were tested to evaluate the consistency and quality of generated results. The AI system produced readable, well-structured, and context-aware content for most test cases.

The generated blog content was displayed instantly within the dashboard interface and stored successfully in the database for future access and editing.

Results Observed

- Blog content generation worked smoothly in real time.
- Generated articles were grammatically correct and properly structured.
- AI-generated titles matched the provided prompts effectively.
- Content history and saved outputs were stored correctly.
- Multiple consecutive generation requests were handled without major performance issues.

The generated outputs demonstrated that the platform could provide useful and practical AI-assisted content creation features for users. Although the system is mainly intended for educational and productivity purposes, the generated blog results showed promising quality, responsiveness, and usability during testing within the *Aether AI* platform.

4.9 AI-Based Image Creation Result

The AI image generation module was tested using different text prompts, image styles, and screen sizes to evaluate both functionality and user experience. The platform successfully generated AI-based images from user prompts in real time while maintaining responsive performance across different devices.

The application adapted properly across:

- Desktop screens
- Laptops
- Mobile devices

Tailwind CSS helped maintain consistent layouts, responsive UI components, and smooth user interaction throughout the image-generation workflow.

Users were able to enter prompts, select image styles, generate images, and preview outputs without interface issues. Generated images were displayed dynamically within the dashboard and stored successfully in cloud storage.

Results Observed

- AI-generated images were created successfully from text prompts.
- Image previews loaded correctly across different screen sizes.
- Navigation between image-generation pages remained smooth.
- Generated outputs reflected quickly on the interface.
- Responsive layouts worked consistently on mobile and desktop devices.
- Multiple image-generation requests were handled without major delays.

The testing results confirmed that the AI image creation module was responsive, stable, and user-friendly, providing a smooth experience for users interacting with Generative AI tools within the *Aether AI* platform.

4.10 Admin Panel – Users, Subscriptions, and Activities

The admin panel was developed to provide centralized control over users, subscriptions, AI tool usage, and overall platform activities within the *Aether AI* platform. The dashboard allowed administrators to monitor platform performance and manage user-related operations efficiently through a secure interface.

The admin panel provided functionalities such as:

- Managing registered users
- Monitoring subscription plans
- Viewing AI tool usage statistics
- Tracking platform activities
- Managing premium feature access
- Monitoring generated content activity

The dashboard displayed important information including:

- Total registered users
- Active subscriptions
- AI content generation statistics
- Recent platform activities
- User engagement and usage data

During testing, administrators were successfully able to access user details, monitor subscription upgrades, and track AI tool activity without performance issues. Role-based access control ensured that only authorized administrators could access the admin dashboard and management features.

Results Observed

- User management operations worked correctly.
- Subscription data updated dynamically after payments.
- Platform activity logs displayed properly.
- AI usage statistics reflected accurately in the dashboard.
- Navigation between admin modules remained smooth and responsive.

The admin panel improved overall platform management by providing a centralized environment for monitoring users, subscriptions, and AI-related activities efficiently within the *Aether AI* ecosystem.

Figure 4.10: Admin Page

4.11 Responsive Home Page of the AI SaaS Platform

Complete end-to-end testing was performed on the responsive home page of the *Aether AI* platform by integrating frontend components, backend APIs, authentication services, database operations, and Generative AI modules together. The homepage was designed to provide users with quick access to all AI-powered tools through a modern and responsive interface.

The following functionalities were tested successfully:

- User registration and login access
- Navigation to AI tools
- AI blog writing access
- AI image generation access

- Background remover access
- Resume review access
- Subscription and upgrade navigation
- Responsive layout behavior across devices

The homepage successfully displayed dynamic platform information such as featured AI tools, usage statistics, subscription details, and recent activities. Navigation between different sections remained smooth, and API responses were reflected quickly within the interface.

The platform adapted properly across:

- Desktop screens
- Laptops
- Tablets
- Mobile devices

Tailwind CSS helped maintain consistent layouts, responsive UI components, and smooth design behavior throughout the application. Images, cards, navigation menus, and dashboard sections adjusted automatically according to screen size without affecting usability.

Results Observed

- Homepage loaded correctly across different devices.
- Navigation between modules remained smooth and responsive.
- Dynamic content updated properly from backend APIs.
- Responsive layouts functioned consistently on smaller screens.
- AI tool access buttons and dashboard sections worked correctly.

The testing confirmed that the responsive home page provided a stable, user-friendly, and visually consistent experience while maintaining smooth communication between all modules of the *Aether AI* platform.

4.12 Cloud Storage and Media Upload Interface

The cloud storage and media upload interface was developed to make image handling simple, fast, and reliable for users of the *Aether AI* platform. Since the platform includes features such as AI image generation and background removal, an efficient media management system was necessary to store and access uploaded and generated files smoothly.

Users were able to upload images directly through the frontend interface without facing complicated steps. Once an image was uploaded, the backend processed the file and securely stored it on Cloudinary cloud storage instead of saving it on the local server. This approach improved performance and reduced server load during testing.

The interface supported multiple image formats including:

- JPEG
- PNG
- WEBP

Uploaded and AI-generated images were displayed properly across different sections of the platform, including the dashboard, recent activity section, generated outputs, and content history pages.

One practical advantage observed during testing was that users could instantly access their uploaded or generated media from anywhere through secure cloud URLs. Images loaded quickly and remained accessible even after refreshing the application or logging in again later.

Results Observed

- Image uploads worked smoothly through the dashboard interface.
- Cloud storage integration functioned correctly.
- Uploaded and generated images displayed properly across the platform.
- Secure image URLs were created automatically.
- Media loading speed remained fast and responsive.
- Different image formats were supported successfully.

Cloudinary also helped optimize images automatically for web delivery, which improved loading speed and overall user experience. The entire upload process felt smooth and natural from the user's perspective, making media handling much more convenient within the *Aether AI* platform.

4.13 PERN Stack Application Deployment Workflow

The deployment workflow of the *Aether AI* platform was designed to ensure that the complete application could run smoothly in a real-world environment. Since the project was developed using the PERN stack (PostgreSQL, Express.js, React.js, and Node.js), proper coordination between the frontend, backend, database, and external AI services was important during deployment.

The deployment process involved configuring the frontend application, backend APIs, PostgreSQL database, cloud storage services, and Generative AI integrations so that all modules could communicate properly after going live.

The overall deployment workflow included:

- Building and optimizing the React frontend
- Hosting backend APIs on the server
- Connecting the PostgreSQL cloud database
- Integrating Cloudinary media storage
- Configuring authentication and environment variables
- Connecting third-party AI APIs for content and image generation

The frontend application was optimized using Vite build tools before deployment, which helped improve loading speed and responsiveness. Backend services handled API requests, authentication, AI processing, and database communication in real time.

During deployment testing, users were successfully able to:

- Access the platform through the browser
- Register and log in securely
- Use AI-powered tools without interruptions
- Generate and save content in real time
- Upload and access media files smoothly

One important observation during deployment was that separating the frontend, backend, and database into independent modules made the application easier to manage and scale. Even when multiple AI requests were performed continuously, the platform remained stable and responsive.

Results Observed

- Frontend and backend communication worked correctly after deployment.
- PostgreSQL database connections remained stable.

- AI APIs responded properly in the deployed environment.
- Cloud-based image storage functioned smoothly.
- The platform remained responsive across different devices and browsers.

Overall, the deployment workflow demonstrated that the *Aether AI* platform could function as a complete real-world AI SaaS application with stable performance, scalable architecture, and smooth integration between all major system components.

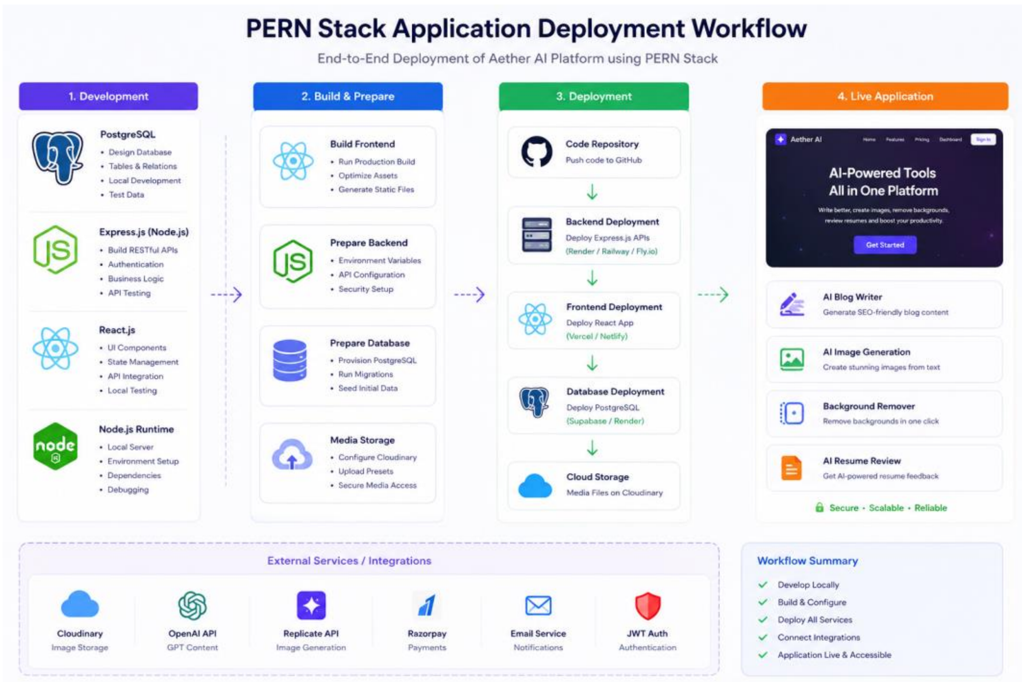

Figure 4.10: Deployment workflow

Chapter 5. Conclusion

5.1 Conclusion

Artificial Intelligence is rapidly transforming the way people create content, manage digital workflows, and interact with online platforms. The idea behind *Aether AI* was to build a smart and user-friendly SaaS platform that brings multiple AI-powered tools together in one place. Instead of using separate applications for writing blogs, generating images, removing backgrounds, or reviewing resumes, users can access all these services through a single integrated platform.

The project successfully combines Generative AI technologies with modern web development practices to create a platform that is practical, responsive, and easy to use. Users can generate AI-written articles, create blog titles, generate images from text prompts, remove image backgrounds, review resumes, and manage their content through a clean dashboard interface.

One of the main strengths of the platform is its simplicity and accessibility. Many AI tools available today can feel complicated or developer-focused, especially for users with limited technical knowledge. In this project, special attention was given to creating an interface that feels smooth, organized, and comfortable to use across different devices.

The integration of multiple AI services within one environment also improved the overall workflow of the application. Instead of switching between different websites and tools, users can complete multiple creative tasks from a single dashboard. Features such as content history, cloud media storage, subscription management, and responsive UI design helped make the platform feel closer to a real-world SaaS application.

From a technical perspective, the project demonstrates how modern technologies like the PERN stack, cloud services, authentication systems, and Generative AI APIs can work together effectively. React.js, Node.js, Express.js, PostgreSQL, Cloudinary, and AI APIs were integrated successfully to create a scalable and modular system architecture.

The platform performed reliably during testing across different modules including authentication, AI content generation, image generation, cloud storage, background removal, subscription handling, and dashboard management. The results showed that the system was capable of handling AI-powered workflows smoothly while maintaining responsive performance and stable communication between all modules.

This project also served as a valuable learning experience by covering different stages of full-stack application development, including frontend development, backend API integration, database management, cloud storage handling, authentication, AI integration, deployment workflows, and responsive UI design.

Overall, *Aether AI* successfully achieved the major objectives defined at the beginning of the project. It demonstrates how Generative AI and modern web technologies can be combined to create intelligent, user-friendly, and scalable SaaS applications that improve productivity and simplify digital content creation.

5.2 Future Works

Although the current version of *Aether AI* includes multiple powerful features, there is still significant scope for future improvements and expansion.

One possible enhancement is the addition of advanced AI models for more personalized and higher-quality content generation. Future versions of the platform could support long-form document generation, AI-powered presentations, code generation, and advanced design assistance tools.

The image-generation system can also be improved further by integrating more advanced text-to-image models and image editing capabilities. Features such as AI video generation, image upscaling, face enhancement, and object editing could make the platform more versatile for creators and businesses.

Another important area for future development is collaboration and team-based workflows. Multi-user workspaces, project sharing, and real-time collaborative editing could help teams use the platform more efficiently for content creation and media management.

The platform can also be extended by integrating:

- Multilingual AI content generation
- AI chatbot assistance
- Voice-to-text and text-to-speech systems
- Social media content automation
- Advanced analytics and user insights
- Mobile application support

Security and scalability can also be improved further through advanced authentication methods, better API rate management, and optimized cloud infrastructure for handling larger user traffic.

At present, the project mainly functions as a prototype and educational AI SaaS platform. With further optimization, cloud deployment improvements, and additional AI integrations, the system could evolve into a production-ready commercial platform in the future.

The growing impact of Generative AI suggests that platforms like *Aether AI* will become increasingly important in digital content creation and productivity workflows. As AI technology continues to advance, integrated SaaS platforms that combine creativity, automation, and accessibility are likely to play a major role in shaping the future of online applications and intelligent digital services.

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